AthletiCore

Train your core

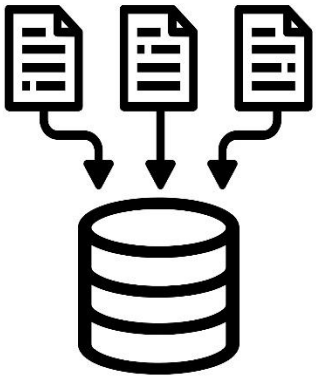
06 MAY 2025

ATHLETICORE SPORTS ACADEMY DATABASE

Group Members : Ayşenur DİKİLİDAŞ
Beraat Can VARLI
Caner ALACALI

Presented to : Vildan ATEŞ

Course : Database Management



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TABLES AND DATA ITEMS

COACH TABLE

- coach_id (Primary Key)
- coach_name
- speciality
- mail
- phone_number

COACH ADDRESS TABLE

- coach_id
- address

STUDENT TABLE

- student_id (Primary Key)
- student_name
- birth
- gender
- mail
- phone_number

STUDENT ADDRESS TABLE

- student_id
- address

ENROLLMENT TABLE

- enrollment_id (Primary Key)
- enrollment_date
- status
- student_id (Foreign Key)
- program_id (Foreign Key)

SPORT TABLE

- sport_id (Primary Key)
- sport_name

PROGRAM TABLE

- program_id (Primary Key)
- program_name
- start_date
- end_date
- coach_id (Foreign Key)
- sport_id (Foreign Key)

PAYMENT TABLE (Weak Entity)

- payment_id (Partial Key)
- enrollment_id
- payment_date
- amount

Requirements-Gathering Method and Business Requirements

Requirements-Gathering Method

The business requirements were determined through one-to-one interviews with coaches at the sports academy and meetings with the consultant faculty member. In addition, the basic needs of the system were revealed by analyzing similar sports academy software.

Business Requirements

- The academy needs a centralized system to manage student and coach information efficiently.
- Personal information and addresses of the students and coaches in the academy should be recorded.
- The system must support tracking of sport programs and ensure that each sport has clear organizational data.
- Students' registration information, the sports they participate in, their classes and their relationship with the responsible coaches should be tracked.
- Each academic term's schedule, including class times and responsible coaches, must be manageable by the system.
- The system must be scalable to support a growing number of students, coaches, and sport branches.
- The platform should minimize manual paperwork by digitizing registrations and attendance.

ENTITIES, ATTRIBUTES, DATA TYPES and CONSTRAINTS

ENTITIES	ATTRIBUTES	DATA TYPE	CONSTRAINTS
STUDENT	student_id	INT	NOT NULL, PRIMARY KEY
	student_name	VARCHAR(100)	
	birth	DATE	
	gender	VARCHAR(10)	
	mail	VARCHAR(100)	
	phone_number	VARCHAR(20)	

STUDENT_ADDRESS	student_id	INT	FOREIGN KEY,(REFERENCES student(student_id)), NOT NULL
	address	VARCHAR(255)	

COACH	coach_id	INT	NOT NULL, PRIMARY KEY
	coach_name	VARCHAR(100)	
	speciality	VARCHAR(100)	
	mail	VARCHAR(100)	
	phone_number	VARCHAR(20)	

COACH_ADDRESS	coach_id	INT	FOREIGN KEY (references coach(coach_id)), DEFAULT NULL
	address	VARCHAR(255)	

SPORT	sport_id	INT	NOT NULL, PRIMARY KEY
	sport_name	VARCHAR(100)	

PROGRAM	program_id	INT	NOT NULL, PRIMARY KEY
	program_name	VARCHAR(100)	
	start_date	DATE	
	end_date	DATE	
	coach_id	INT	FOREIGN KEY (references coach(coach_id)), DEFAULT NULL
	sport_id	INT	FOREIGN KEY (references sport(sport_id)), DEFAULT NULL

ENROLLMENT	enrollment_id	INT	NOT NULL, PRIMARY KEY
	enrollment_date	DATE	
	status	VARCHAR(50)	
	student_id	INT	FOREIGN KEY (references student(student_id)), DEFAULT NULL
	program_id	INT	FOREIGN KEY (references program(program_id)), DEFAULT NULL

PAYMENT	payment_id	INT	
	enrollment_id	INT	PRIMARY KEY, FOREIGN KEY (references enrollment(enrollment_id))
	payment_date	DATE	
	amount	DECIMAL(10,2)	

Note:

The tables of the coach_address and student_address entities were created after normalization.

Detailed information will be provided later on in the normalization topic.

RELATIONSHIPS BETWEEN ENTITIES and TYPES

1- Coach ↔ Program (1:N)

- **Type:** One-to-many relationship
- *A program can only be managed by one coach, but a coach may manage multiple programs.*

2- Sport ↔ Program (1:N)

- Type: One-to-many relationship
- *A sport can have multiple programs, but each program focuses on only one sport.*

3- Student ↔ Enrollment (1:N)

- **Type:** One-to-many relationship
- *A student can enroll in multiple programs, but each enrollment belongs to only one student.*

4- Program ↔ Enrollment (1:N)

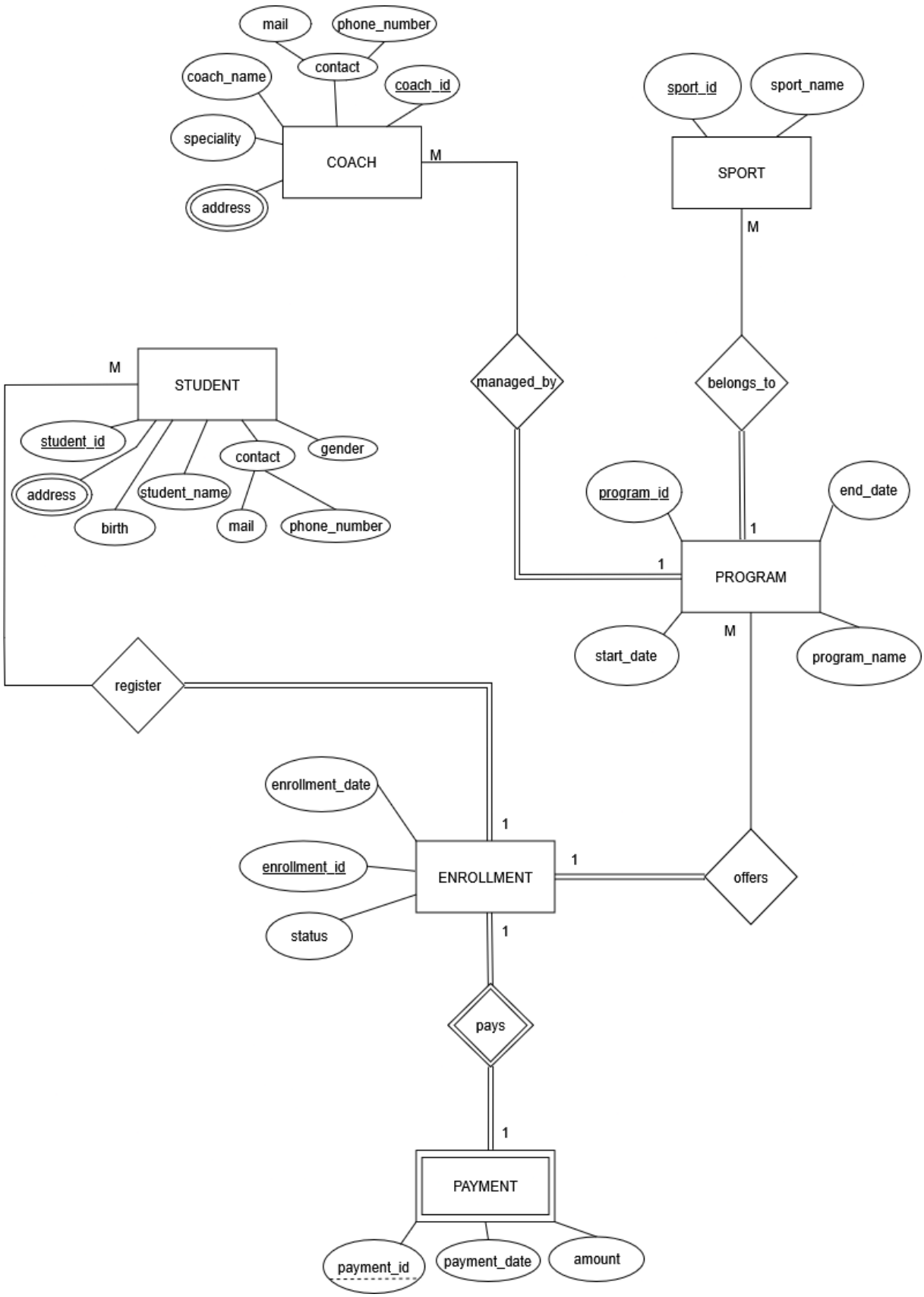
- **Type:** One-to-many relationship
- *A program can have multiple student enrollments, but each enrollment is for only one program.*

5- Enrollment ↔ Payment (1:1)

- **Type:** One-to-one relationship
- *An enrollment can have only one payment, and each payment corresponds to exactly one enrollment.*
- *In this relationship, since the payment entity is a weak entity that depends on enrollment, the relationship type between them is identifying*.

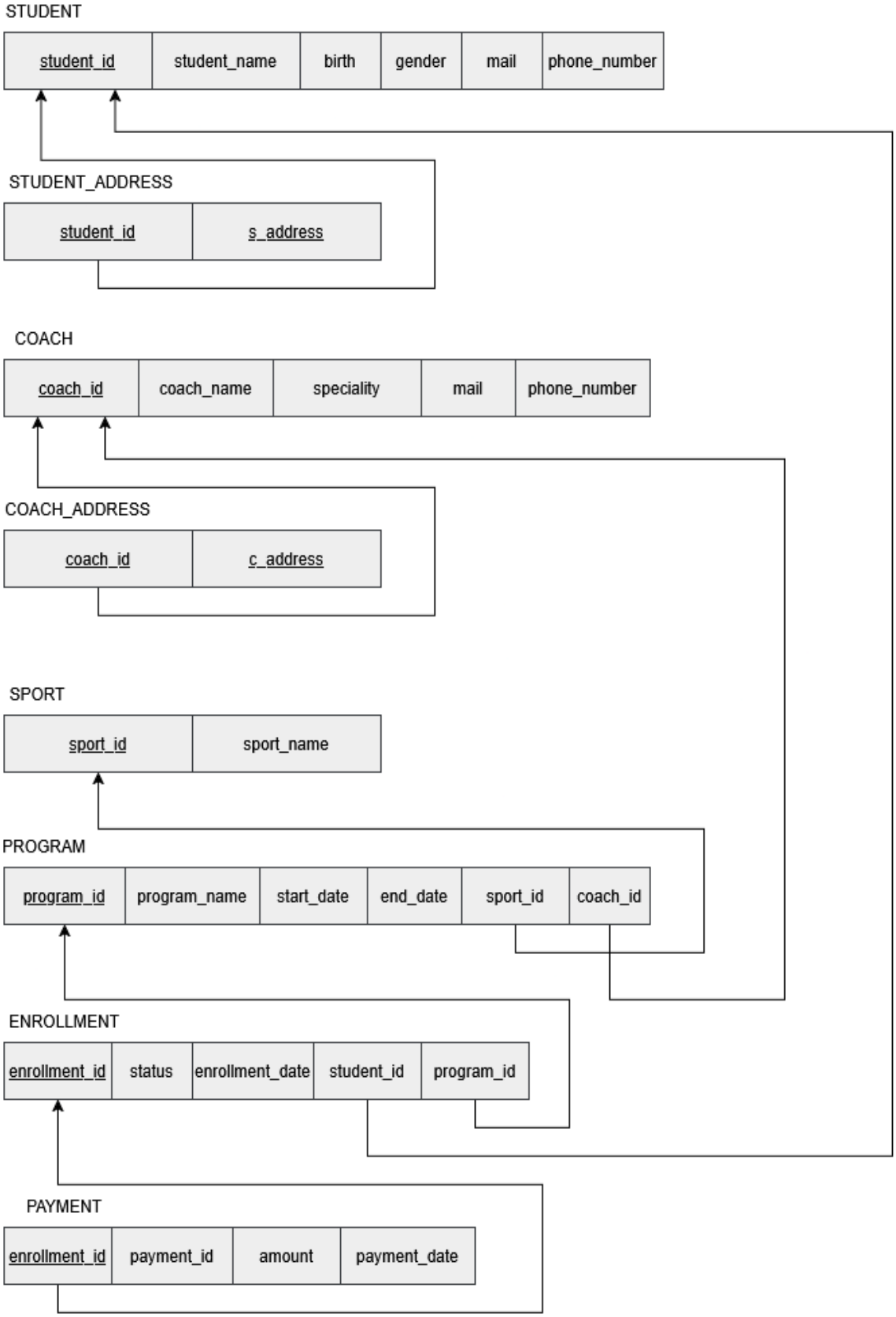
ER DIAGRAM OF

ATHLETICORE SPORTS ACADEMY DATABASE



RELATIONAL MAPPING PROCESS OF

ATHLETICORE SPORTS ACADEMY DATABASE



CREATING TABLES

```sql
CREATE TABLE coach (
    coach_id INT NOT NULL PRIMARY KEY,
    coach_name VARCHAR(100),
    speciality VARCHAR(100),
    mail VARCHAR(100),
    phone_number VARCHAR(20),
    address VARCHAR(80)
);
```

```sql
CREATE TABLE student (
    student_id INT NOT NULL PRIMARY KEY,
    student_name VARCHAR(100),
    birth DATE,
    gender VARCHAR(10),
    mail VARCHAR(100),
    phone_number VARCHAR(20),
    address VARCHAR(80)
);
```

```sql
CREATE TABLE sport (
    sport_id INT NOT NULL PRIMARY KEY,
    sport_name VARCHAR(100)
);
```

```sql
CREATE TABLE program (
  program_id INT NOT NULL PRIMARY KEY,
  program_name VARCHAR(100),
  start_date DATE,
  end_date DATE,
  coach_id INT,
  sport_id INT,
  FOREIGN KEY (coach_id) REFERENCES coach(coach_id),
  FOREIGN KEY (sport_id) REFERENCES sport(sport_id)
);
```

```sql
CREATE TABLE enrollment (
  enrollment_id INT NOT NULL PRIMARY KEY,
  enrollment_date DATE,
  status VARCHAR(50),
  student_id INT,
  program_id INT,
  FOREIGN KEY (student_id) REFERENCES student(student_id),
  FOREIGN KEY (program_id) REFERENCES program(program_id)
);
```

```sql
CREATE TABLE payment (
  enrollment_id INT NOT NULL PRIMARY KEY,
  payment_id INT,
  payment_date DATE,
  amount DECIMAL(10,2),
  FOREIGN KEY (enrollment_id) REFERENCES enrollment(enrollment_id)
);
```

INSERTING VALUES ON TABLES

```
INSERT INTO `coach` VALUES
(1,'Murat Akgündüz','Football','murat.akgunduz@sporakademi.com','55087859645','Günçiçek Mahallesi Fidan Caddesi, 36077.Sokak Menekşe Apartmanı, Kırıkkale'),
(2,'Kerem Yalın Şener','Basketball','yalin.sener@sporakademi.com','51618923120','Altgeçit Mahallesi Papatya Caddesi, 873. Sokak Gül Apartmanı, Ankara'),
(3, 'Asuman Uğurlu', 'Swimming', 'asuman.ugurlu@sporakademi.com', '53312577892',
 'Altıntepe Mahallesi, 773. Sokak, Ankara; Çiçek Mahallesi, 456. Sokak, Kırıkkale'),
(4,'Ferhat Özçelik','Tennis','ferhat.ozcelik@sporakademi.com','59900122345','Bahçelievler Mahallesi, 447. Sokak, Ankara'),
(5, 'Melda Koçak', 'Volleyball', 'melda.kocak@sporakademi.com', '50833991456',
 'Kurtuluş Mahallesi, 123. Sokak, Kırıkkale; Barış Mahallesi, 888. Sokak, Ankara');
```

```
INSERT INTO `student` VALUES
(1, 'Erik Şensoy', '2007-06-15', 'Male', 'ifirat@yahoo.com', '56681499556',
 'Türk Mahallesi, 979. Sokak, Ankara; Ahi Mesut Mahallesi, 456. Sokak, Ankara'),
(2,'Sümerya Akçay Yorulmaz','2012-06-17','Male','cangur41@hotmail.com','55515133975','Soykut Soylu Mahallesi, 49511. Sokak, Ankara'),
(3,'Hilal Yurtçu','2007-04-14','Female','lkarabulut@hotmail.com','56500411592','Elifce Mahallesi, 636. Sokak, Kırıkkale'),
(4, 'Kurt Torun', '2011-01-24', 'Male', 'azimdemirel@kaplanlar.com', '50823967129',
 'Avci Kent Mahallesi, 788. Sokak, Ankara; Zafer Mahallesi, 222. Sokak, Kırıkkale'),
(5,'Fikriye Kula','2012-08-23','Female','umitdemir@hotmail.com','59829347020','Sevimli Vadisi Mahallesi, 847. Sokak, Ankara'),
(6,'Yusuf Emre Sütçü','2011-09-19','Male','oktayilhan@yahoo.com','56129740002','Mertçe Bahçesi Mahallesi, 773. Sokak, Ankara'),
(7, 'Turan Kızıltepe', '2009-10-24', 'Male', 'akarsu.ipek@terlikci.com', '54422702139',
 'Göksu Yaylası Mahallesi, 609. Sokak, Kırıkkale; Kızılırmak Mahallesi, 345. Sokak, Ankara'),
(8,'Sevgi Alper','2007-12-05','Female','elifdaglar@hotmail.com','58212539384','Yeşilkent Mahallesi, 843. Sokak, Ankara'),
(9,'Gülten Kutluer','2013-02-15','Female','fatmagul@gmail.com','53500639114','Çamlık Tepesi Mahallesi, 231. Sokak, Ankara'),
(10,'Hüseyin Şeker','2011-03-21','Male','m.seker@posta.com','53682537845','Gölbaşı Mahallesi, 714. Sokak, Ankara');
```

```
INSERT INTO `sport` VALUES
(1,'Football'),
(2,'Basketball'),
(3,'Volleyball'),
(4,'Swimming'),
(5,'Tennis'),
(6,'Athletics'),
(7,'Handball'),
(8,'Archery');
```

```sql
INSERT INTO `program` VALUES
(1,'Football Development Program','2025-03-01','2025-05-31',1,1),
(2,'Basketball Basic Training','2025-04-10','2025-06-10',2,2),
(3,'Swimming Technique Development','2025-05-05','2025-07-05',3,4),
(4,'Tennis Advanced Program','2025-06-01','2025-08-01',4,5),
(5,'Volleyball Skills Enhancement','2025-04-20','2025-06-20',5,3);
```

```sql
INSERT INTO `enrollment` VALUES
(1,'2025-04-20','Pending',1,1),
(2,'2025-04-22','Completed',2,1),
(3,'2025-04-21','Pending',3,2),
(4,'2025-04-25','Completed',4,3),
(5,'2025-04-23','Completed',5,4),
(6,'2025-04-24','Pending',6,2),
(7,'2025-04-22','Pending',7,5),
(8,'2025-04-21','Completed',8,3),
(9,'2025-04-20','Completed',9,4),
```

```sql
INSERT INTO `payment` VALUES
(1,1001,'2025-04-28',1000.00),
(2,1002,'2025-04-27',800.00),
(3,1003,'2025-04-29',950.00),
(4,1004,'2025-04-30',1200.00),
(5,1005,'2025-04-26',1100.00),
(6,1006,'2025-04-28',1050.00),
(7,1007,'2025-04-29',1000.00),
(8,1008,'2025-04-27',1150.00),
(9,1009,'2025-04-30',900.00),
```

NORMALIZATION PROCESS

BEFORE NORMALIZATION

Coach Table

coach_id	coach_name	address
1	Murat Akgündüz	Günçiçek Mahallesi Fidan Caddesi, 36077.Sokak Menekşe Apartmanı, Kırıkkale
2	Kerem Yalın Şener	Altgeçit Mahallesi Papatya Caddesi, 873. Sokak Gül Apartmanı, Ankara
3	Asuman Uğurlu	Altıntepe Mahallesi, 773. Sokak, Ankara ; Çiçek Mahallesi, 456. Sokak, Kırıkkale
4	Ferhat Özçelik	Bahçelievler Mahallesi, 447. Sokak, Ankara
5	Melda Koçak	Kurtuluş Mahallesi, 123. Sokak, Kırıkkale ; Barış Mahallesi, 888. Sokak, Ankara

Student Table

student_id	student_name	address
1	Erik Şensoy	Türk Mahallesi, 979. Sokak, Ankara; Ahi Mesut Mahallesi, 456. Sokak, Ankara
2	Sümerya Akçay Yorulmaz	Soykut Soylu Mahallesi, 49511. Sokak, Ankara
3	Hilal Yurtçu	Elifce Mahallesi, 636. Sokak, Kırıkkale
4	Kurt Torun	Avci Kent Mahallesi, 788. Sokak, Ankara; Zafer Mahallesi, 222. Sokak, Kırıkkale
5	Fikriye Kula	Sevimli Vadisi Mahallesi, 847. Sokak, Ankara
6	Yusuf Emre Sütçü	Mertçe Bahçesi Mahallesi, 773. Sokak, Ankara
7	Turan Kızıltepe	Gök Mahallesi, 609. Sokak, Kırıkkale; Kızıl Mahallesi, 345. Sokak, Ankara
8	Sevgi Alper	Yeşilkent Mahallesi, 843. Sokak, Ankara
9	Gülten Kutluer	Çamlık Tepesi Mahallesi, 231. Sokak, Ankara
10	Hüseyin Şeker	Gölbaşı Mahallesi, 714. Sokak, Ankara

- ✓ *In the tables above, students and coaches with more than one address in their address in some sections(coach_id = 3 and 5, student_id = 1, 4 and 7) do not comply with the first normal form (1NF). To prevent this and to get through the normalization process without any problems, we need to create separate tables for these two entities.*

AFTER NORMALIZATION

Coach Table

```sql
CREATE TABLE coach_address (
    coach_id INT,
    address VARCHAR(255),
    FOREIGN KEY (coach_id) REFERENCES coach(coach_id)
);
```

```sql
INSERT INTO coach_address
VALUES
(1,'Günçiçek Mahallesi Fidan Caddesi, 36077.Sokak Menekşe Apartmanı, Kırıkkale,'),
(2,'Altgeçit Mahallesi Papatya Caddesi, 873. Sokak Gül Apartmanı, Ankara'),
(3,'Altıntepe Mahallesi, 773. Sokak, Ankara'),
(3,'Çiçek Mahallesi, 456. Sokak, Kırıkkale'),
(4,'Bahçelievler Mahallesi, 447. Sokak, Ankara'),
(5,'Kurtuluş Mahallesi, 123. Sokak, Kırıkkale');
(5,'Barış Mahallesi, 888. Sokak, Ankara');
```

coach_id	address
1	Günçiçek Mahallesi Fidan Caddesi, 36077.Sokak Menekşe Apartmanı, Kırıkkale
2	Altgeçit Mahallesi Papatya Caddesi, 873. Sokak Gül Apartmanı, Ankara
3	Altıntepe Mahallesi, 773. Sokak, Ankara
3	Çiçek Mahallesi, 456. Sokak, Kırıkkale
4	Bahçelievler Mahallesi, 447. Sokak, Ankara
5	Kurtuluş Mahallesi, 123. Sokak, Kırıkkale
5	Barış Mahallesi, 888. Sokak, Ankara

- ✓ The table named coach_address was created because the repeated addresses in the coach table did not conform to the first normal form(1NF). All coaches' addresses were entered into the tables. Since the addresses of coaches with coach_id 3 and 5 were repeated, separate rows were opened for each coach's different addresses and their values were entered.

Student Table

```sql
CREATE TABLE student_address (
    student_id INT,
    address VARCHAR(255),
    FOREIGN KEY (student_id) REFERENCES student(student_id)
);
```

```sql
INSERT INTO student_address (student_id, address)
VALUES
(1,'Türk Mahallesi, 979. Sokak, Ankara'),
(1, 'Ahi Mesut Mahallesi, 456. Sokak, Ankara'),
(2,'Soykut Soylu Mahallesi, 49511. Sokak, Ankara'),
(3,'Elifce Mahallesi, 636. Sokak, Kırıkkale'),
(4,'Avci Kent Mahallesi, 788. Sokak, Ankara'),
(4, 'Zafer Mahallesi, 222. Sokak, Kırıkkale'),
(5,'Sevimli Vadisi Mahallesi, 847. Sokak, Ankara'),
(6,'Mertçe Bahçesi Mahallesi, 773. Sokak, Ankara'),
(7,'Göksu Yaylası Mahallesi, 609. Sokak, Kırıkkale'),
(7, 'Kızılırmak Mahallesi, 345. Sokak, Ankara'),
(8,'Yeşilkent Mahallesi, 843. Sokak, Ankara'),
```

student_id	address
1	Türk Mahallesi, 979. Sokak, Ankara
1	Ahi Mesut Mahallesi, 456. Sokak, Ankara
2	Soykut Soylu Mahallesi, 49511. Sokak, Ankara
3	Elifce Mahallesi, 636. Sokak, Kırıkkale
4	Avci Kent Mahallesi, 788. Sokak, Ankara
4	Zafer Mahallesi, 222. Sokak, Kırıkkale
5	Sevimli Vadisi Mahallesi, 847. Sokak, Ankara
6	Mertçe Bahçesi Mahallesi, 773. Sokak, Ankara
7	Göksu Yaylası Mahallesi, 609. Sokak, Kırıkkale
7	Kızılırmak Mahallesi, 345. Sokak, Ankara
8	Yeşilkent Mahallesi, 843. Sokak, Ankara

✓ The table named student_address was created because the repeated addresses in the student table did not conform to the first normal form(1NF). All students' addresses were entered into the tables. Since the addresses of students with student_id 1, 4 and 7 were repeated, separate rows were opened for each student's different addresses and their values were entered.

Our tables now conform to all forms of normalization (1NF, 2NF and 3NF).

We can explain how they conform to 2nd and 3rd normal forms as follows:

Compliance with 2NF (Second Normal Form)

2NF Rules:

1. Must satisfy 1NF (all attributes contain atomic values).
2. No partial dependencies (non-prime attributes must depend on the *entire* composite PK, if one exists).

Proof of 2NF Compliance:

- All tables satisfy 1NF. No tables contain **repeating groups** or **non-atomic values**.
- No partial dependencies.
- No composite primary keys exist. Automatically satisfy 2NF.

Compliance with 3NF (Third Normal Form)

3NF Rules:

1. Must satisfy 2 NF.
2. No transitive dependencies.

Proof of 3NF Compliance:

- When transitive dependency checks were checked, no incompatibilities were found.
- No table has a non-prime attribute dependent on another non-prime attribute.

SQL QUERY EXAMPLES

Query 1:

```
SELECT student_name, mail
FROM student;
```

- Prints the student_name and e-mail information from the student table to the screen.

student_name	mail
Erik Şensoy	ifirat@yahoo.com
Sümerya Akçay Yorulmaz	cangur41@hotmail.com
Hilal Yurtçu	lkarabulut@hotmail.com
Kurt Torun	azimdemirel@kaplanlar.com
Fikriye Kula	umitdemir@hotmail.com
Yusuf Emre Sütçü	oktayilhan@yahoo.com
Turan Kızıltepe	akarsu.ipek@terlikci.com
Sevgi Alper	elifdaglar@hotmail.com
Gülten Kutluer	fatmagul@gmail.com
Hüseyin Şeker	m.seker@posta.com
Ahmet Üstün	ahmetustun@yahoo.com
Ayşenur Akkaya	akkaya_aysenur@posta.com
Mustafa Güler	mguler@hotmail.com
Elif Nur Kaya	elif.kaya@gmail.com
Emre Uysal	emreuysal@posta.com
Meryem Koç	meryemkoc@hotmail.com

Query 2:

```
SELECT student_name, birth
FROM student
WHERE birth < '2005.01.01';
```

- Shows the names and birth dates of students whose birth date is greater than 2005.01.01 from the student table.

student_name	birth
Serhat Taş	2001-12-01
Duygu Karaca	2002-05-15
Onur Özkan	2004-09-05

Query 3:

```
SELECT gender, COUNT(*) AS total
FROM student
GROUP BY gender;
```

gender	total
Male	16
Female	14

- This SQL query counts and groups the number of students by their gender.

Query 4:

```sql
SELECT student_name, birth
FROM student
ORDER BY birth DESC
LIMIT 5;
```

- This SQL query retrieves the names and birth dates of the 5 youngest students from the student table.

student_name	birth
Beyza Ekinci	2013-03-09
Gülten Kutluer	2013-02-15
Fikriye Kula	2012-08-23
Sümerya Akçay Yorulmaz	2012-06-17
Meryem Koç	2012-02-12

Query 5:

```sql
SELECT SUM(amount) AS total_income
FROM payment;
```

total_income
30850.00

- This SQL query calculates and returns the total sum of all payment amounts as "total_income".

Query 6:

```sql
SELECT program_name, start_date
FROM program
WHERE start_date > CURDATE();
```

- This SQL query retrieves the names and start dates of all programs that are scheduled to begin after the current date.

program_name	start_date
Swimming Technique Development	2025-05-05
Tennis Advanced Program	2025-06-01

Query 7:

```sql
SELECT program_name, DATEDIFF(end_date, start_date) AS duration_days
FROM program
WHERE DATEDIFF(end_date, start_date) > 30;
```

- This SQL query lists program names along with their duration in days, showing only programs that last more than 30 days.

program_name	duration_days
Football Development Program	91
Basketball Basic Training	61
Swimming Technique Development	61
Tennis Advanced Program	61
Volleyball Skills Enhancement	61

Query 8:

```
SELECT YEAR(birth) AS birth_year, COUNT(*) AS student_count
FROM student
GROUP BY birth_year;
```

birth_year	student_count
2007	6
2012	3
2011	5
2009	4
2013	2
2008	4
2010	3
2001	1
2002	1
2004	1

- This SQL query counts how many students were born in each unique year and displays the results as **birth_year** and **student_count**.

Query 9:

```sql
SELECT p.program_name, c.coach_name
FROM program p
INNER JOIN coach c ON p.coach_id = c.coach_id;
```

- This SQL query retrieves a list of program names along with the corresponding coach names assigned to each program.

program_name	coach_name
Football Development Program	Murat Akgündüz
Basketball Basic Training	Kerem Yalın Şener
Swimming Technique Development	Asuman Uğurlu
Tennis Advanced Program	Ferhat Özçelik
Volleyball Skills Enhancement	Melda Koçak

Query 10:

```sql
SELECT pr.program_name, COUNT(e.enrollment_id) AS num_students
FROM program pr
LEFT JOIN enrollment e ON pr.program_id = e.program_id
GROUP BY pr.program_name;
```

- This SQL query calculates the number of students enrolled in each program and displays the results.

program_name	num_students
Football Development Program	7
Basketball Basic Training	6
Swimming Technique Development	6
Tennis Advanced Program	6
Volleyball Skills Enhancement	5

Query 11:

```sql
SELECT s.student_name, sa.address
FROM student s
LEFT JOIN student_address sa ON s.student_id = sa.student_id
WHERE sa.address LIKE '%Kırıkkale';
```

- This SQL query retrieves student names along with their addresses, specifically filtering for students who have addresses in **Kırıkkale**.

student_name	address
Hilal Yurtçu	Elifce Mahallesi, 636. Sokak, Kırıkkale
Kurt Torun	Zafer Mahallesi, 222. Sokak, Kırıkkale
Turan Kızıltepe	Göksu Yaylası Mahallesi, 609. Sokak, Kırıkkale
Ahmet Üstün	Yenimahalle Mahallesi, 311. Sokak, Kırıkkale
Elif Nur Kaya	İncesu Mahallesi, 284. Sokak, Kırıkkale
İsmail Çetin	Bahçelievler Mahallesi, 301. Sokak, Kırıkkale
Okan Yıldız	Kocatepe Mahallesi, 610. Sokak, Kırıkkale
Halil İbrahim Aslan	Gölbaşı Mahallesi, 847. Sokak, Kırıkkale

Query 12:

```
SELECT s.student_name, e.enrollment_date
FROM enrollment e
INNER JOIN student s ON e.student_id = s.student_id
WHERE e.status = "Completed";
```

- This SQL query retrieves the names of students and their enrollment dates for all **successfully completed** enrollments.

student_name	enrollment_date
Sümerya Akçay Yorulmaz	2025-04-22
Kurt Torun	2025-04-25
Fikriye Kula	2025-04-23
Sevgi Alper	2025-04-21
Gülten Kutluer	2025-04-20
Ayşenur Akkaya	2025-04-22
Emre Uysal	2025-04-21
Meryem Koç	2025-04-22
Kerem Polat	2025-04-21
Okan Yıldız	2025-04-23
Burak Tan	2025-04-25
Ebru Aydın	2025-04-22
Duygu Karaca	2025-04-21
Gizem Arslan	2025-04-22

Query 13:

```sql
SELECT s.student_name, SUM(p.amount) AS total_paid
FROM payment p
INNER JOIN enrollment e ON p.enrollment_id = e.enrollment_id
INNER JOIN student s ON e.student_id = s.student_id
GROUP BY s.student_name
ORDER BY total_paid DESC
LIMIT 5;
```

- This SQL query identifies the **top 5 highest-paying students** by calculating the total amount each student has paid across all their enrollments.

student_name	total_paid
Ebru Aydın	1200.00
Mustafa Güler	1200.00
Kurt Torun	1200.00
Sevgi Alper	1150.00
Seda Yılmaz	1150.00

Query 14:

```sql
SELECT sp.sport_name, SUM(p.amount) AS total_income
FROM sport sp
JOIN program pr ON sp.sport_id = pr.sport_id
JOIN enrollment e ON pr.program_id = e.program_id
JOIN payment p ON e.enrollment_id = p.enrollment_id
GROUP BY sp.sport_name;
```

- This SQL query calculates the **total revenue generated per sport** by aggregating payment amounts across all related programs and enrollments.

sport_name	total_income
Football	7200.00
Basketball	5750.00
Volleyball	5100.00
Swimming	6450.00
Tennis	6350.00

Query 15:

```sql
SELECT s.student_name, pr.program_name, sp.sport_name
FROM enrollment e
INNER JOIN student s ON e.student_id = s.student_id
INNER JOIN program pr ON e.program_id = pr.program_id
INNER JOIN sport sp ON pr.sport_id = sp.sport_id;
```

- This SQL query generates a comprehensive **student-program-sport relationship report**, showing which students are enrolled in which programs and the associated sports.

student_name	program_name	sport_name
Erik Şensoy	Football Development Program	Football
Sümerya Akçay Yorulmaz	Football Development Program	Football
Hüseyin Şeker	Football Development Program	Football
Ayşenur Akkaya	Football Development Program	Football
İsmail Çetin	Football Development Program	Football
Okan Yıldız	Football Development Program	Football
Ebru Aydın	Football Development Program	Football
Hilal Yurtçu	Basketball Basic Training	Basketball
Yusuf Emre Sütçü	Basketball Basic Training	Basketball
Ahmet Üstün	Basketball Basic Training	Basketball
Meryem Koç	Basketball Basic Training	Basketball
Zeynep Güneş	Basketball Basic Training	Basketball
Serhat Taş	Basketball Basic Training	Basketball
Kurt Torun	Swimming Technique Development	Swimming
Sevgi Alper	Swimming Technique Development	Swimming
Mustafa Güler	Swimming Technique Development	Swimming
Beyza Ekinci	Swimming Technique Development	Swimming
Burak Tan	Swimming Technique Development	Swimming
Duygu Karaca	Swimming Technique Development	Swimming
Fikriye Kula	Tennis Advanced Program	Tennis
Gülten Kutluer	Tennis Advanced Program	Tennis
Elif Nur Kaya	Tennis Advanced Program	Tennis
Kerem Polat	Tennis Advanced Program	Tennis
Seda Yılmaz	Tennis Advanced Program	Tennis
Onur Özkan	Tennis Advanced Program	Tennis
Turan Kızıltepe	Volleyball Skills Enhancement	Volleyball
Emre Uysal	Volleyball Skills Enhancement	Volleyball
Melike Demir	Volleyball Skills Enhancement	Volleyball
Halil İbrahim Aslan	Volleyball Skills Enhancement	Volleyball
Gizem Arslan	Volleyball Skills Enhancement	Volleyball